

# Supplementary Material

## An age-structured contact model of sick-leave and school-absence effects on seasonal influenza in Korea

This document provides supplementary tables and figures referenced in the main text: season-specific population data and the robustness of the calibration to the population input (Section S1); the contact structure (Section S2); the age-resolved model fit (Section S3); baseline epidemic dynamics (Section S4); age-specific policy effects across all three seasons (Section S5); the policy dose-response (Section S6); and additional sensitivity analyses (Section S7).

### **S1 Population data and robustness to the population input**

The model uses season-specific resident-registration populations for the Seoul Capital Area (Seoul, Incheon, and Gyeonggi), taken from the Statistics Korea (KOSIS) table of resident-registration population by administrative district (*eup/myeon/dong*) and 5-year age band, aggregated to the 15 model age bands (with 70+ the sum of 70–74 through 100+) and to the three provinces. Each season uses the population of June of its peak year: 2016 for 2016–17, 2017 for 2017–18, and 2019 for 2019–20 (Table S1). The declining child and rising elderly shares across seasons reflect the population aging of the study area.

Table S1: Season-specific resident-registration population by age band (Seoul, Incheon, and Gyeonggi combined; June of each peak year).

| Age band     | 2016–17 (Jun 2016) | 2017–18 (Jun 2017) | 2019–20 (Jun 2019) |
|--------------|--------------------|--------------------|--------------------|
| 0-4          | 1,105,037          | 1,042,254          | 933,536            |
| 5-9          | 1,176,897          | 1,172,633          | 1,156,547          |
| 10-14        | 1,164,698          | 1,167,674          | 1,146,612          |
| 15-19        | 1,520,515          | 1,439,181          | 1,311,672          |
| 20-24        | 1,783,137          | 1,755,477          | 1,689,028          |
| 25-29        | 1,753,436          | 1,822,973          | 1,945,898          |
| 30-34        | 1,905,403          | 1,807,548          | 1,761,455          |
| 35-39        | 2,121,660          | 2,146,295          | 2,095,520          |
| 40-44        | 2,181,750          | 2,119,022          | 2,028,568          |
| 45-49        | 2,321,126          | 2,349,511          | 2,312,574          |
| 50-54        | 2,069,910          | 2,058,334          | 2,164,205          |
| 55-59        | 2,028,672          | 2,080,066          | 2,091,185          |
| 60-64        | 1,463,595          | 1,546,135          | 1,774,446          |
| 65-69        | 997,193            | 1,048,866          | 1,153,998          |
| 70+          | 1,997,436          | 2,123,894          | 2,360,555          |
| <b>Total</b> | 25,590,465         | 25,679,863         | 25,925,799         |

Replacing a single reference-year population (January 2023) with these year-specific populations, and re-fitting, leaves the calibration essentially unchanged (Table S2): the channel allocation shifts by at most 0.004 per channel, season-specific  $R_0$  by less than 1%, and no  $0 \notin \text{CI}$  status changes. The adult-down / school-age-up redistribution direction and the school-over-sick-leave ordering held in all three seasons under both population inputs.

Table S2: Posterior comparison under a single reference-year population (2023) versus season-specific populations. Values are posterior means; channel shares are shared across seasons,  $R_0$  is season-specific.

| Parameter             | Single-year (2023) | Season-specific | Change |
|-----------------------|--------------------|-----------------|--------|
| $\pi_{\text{home}}$   | 0.445              | 0.442           | −0.003 |
| $\pi_{\text{work}}$   | 0.294              | 0.298           | +0.004 |
| $\pi_{\text{school}}$ | 0.065              | 0.065           | 0.000  |
| $\pi_{\text{other}}$  | 0.195              | 0.195           | 0.000  |
| $R_0$ (2016–17)       | 2.276              | 2.287           | +0.5%  |
| $R_0$ (2017–18)       | 2.071              | 2.078           | +0.3%  |
| $R_0$ (2019–20)       | 2.164              | 2.169           | +0.3%  |

## S2 Contact structure

Contacts are represented by four channel-specific matrices (home, work, school, other), each  $15 \times 15$  over the age bands, measured separately for school term and winter break

by the NIMS contact survey (main text). The time-varying matrix  $C(t)$  blends the two by a trapezoidal weight  $h(t)$  that rises over the winter break: transition onset at day 113, plateau over days 127–162, and return by day 183. Figure S1 shows the school-term and winter-break matrices for each channel; the marked reduction in the school block during the winter break is what the calendar switch encodes.

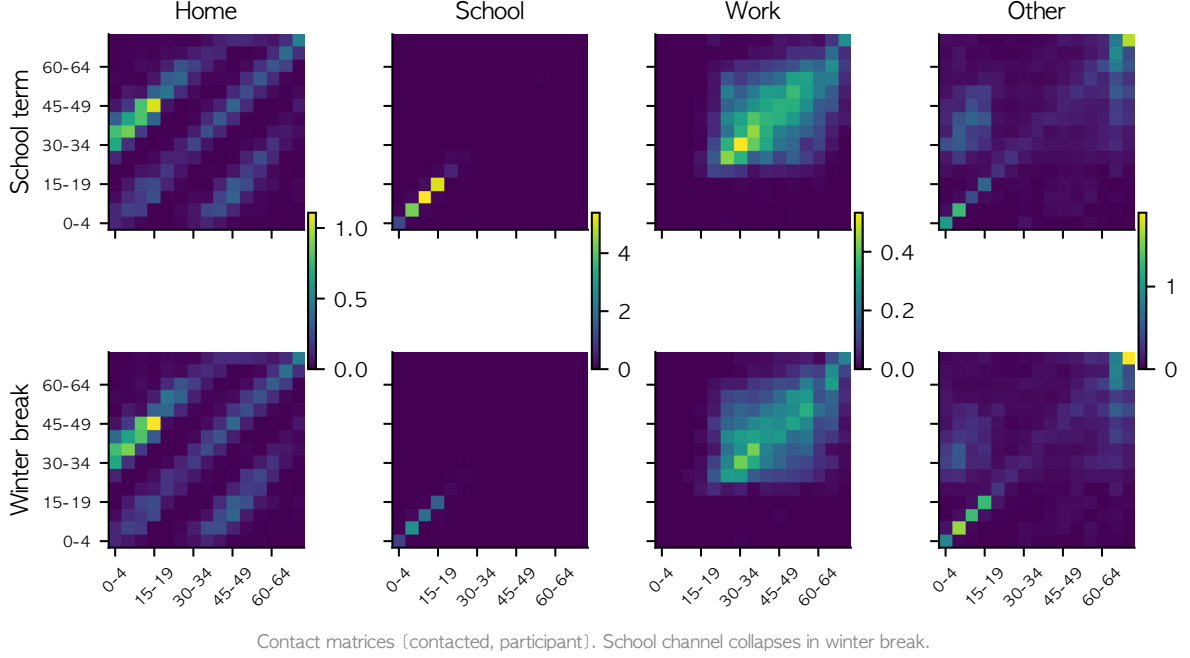


Figure S1: Age-structured contact matrices by channel (home, work, school, other), school term versus the winter break.

### S3 Age-resolved model fit

Figure S2 shows the observed NHIS incidence and the posterior model by age band for each season, complementing the aggregate fit in the main text. The over-prediction of epidemic width discussed in the main text is most visible in the sharp 2016–17 season and in the youngest band.

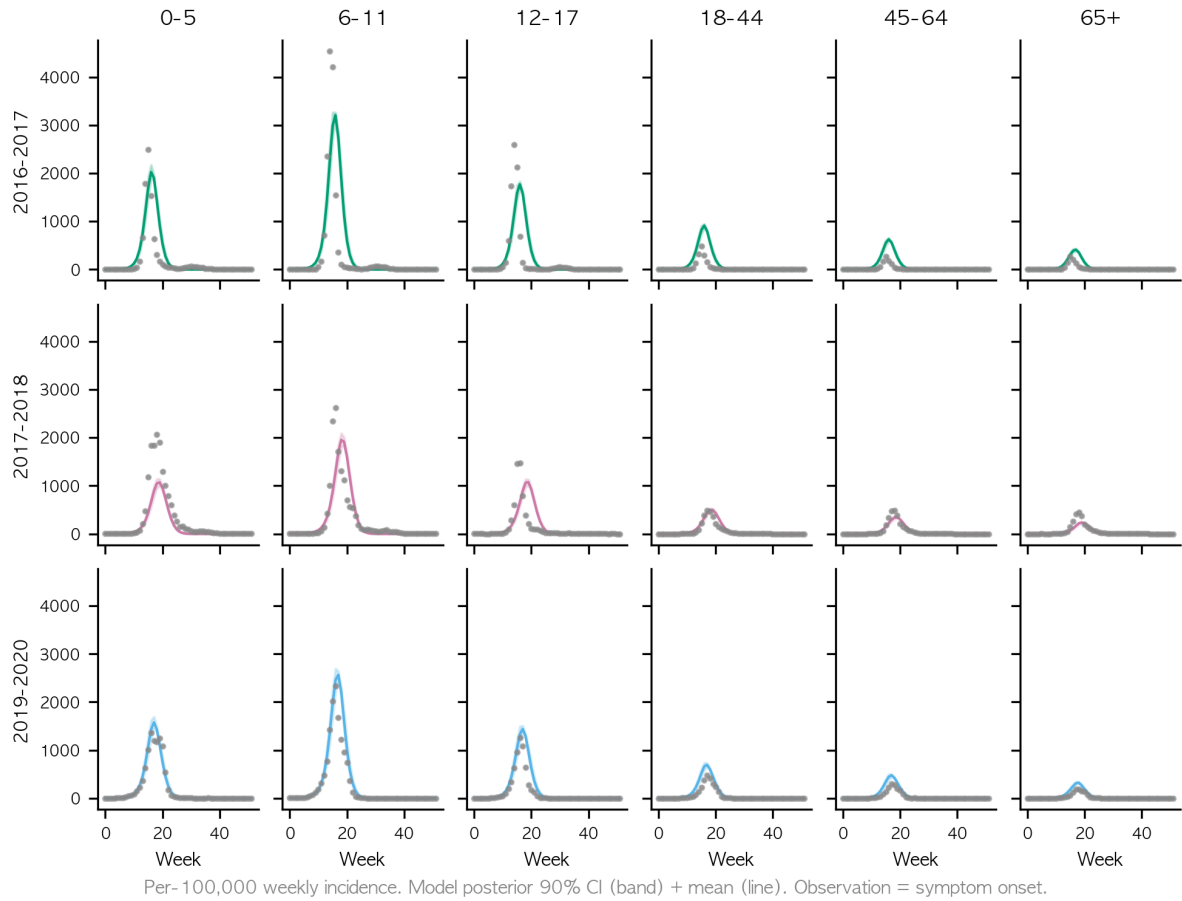


Figure S2: Observed (points) versus posterior model (line with 90% credible band) incidence by age band and season.

## S4 Baseline epidemic dynamics

Figure S3 shows the modelled age-specific incidence over the season at baseline (no intervention), and Figure S4 the baseline final attack rate by age. School-age children carry the highest attack rates and peak early, consistent with the school channel acting as the epidemic engine.

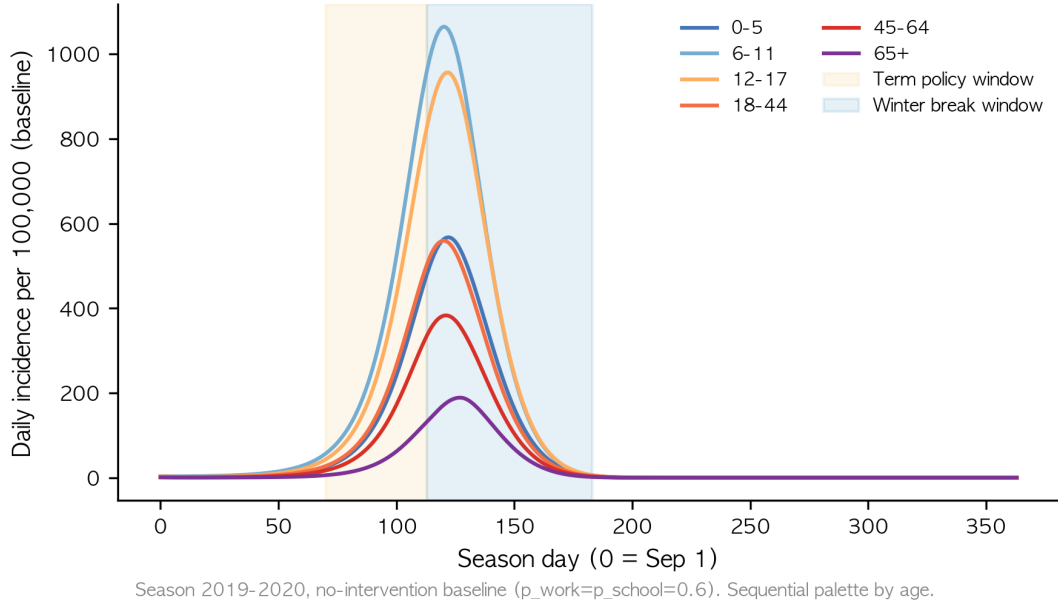


Figure S3: Modelled age-specific epidemic curves at baseline.

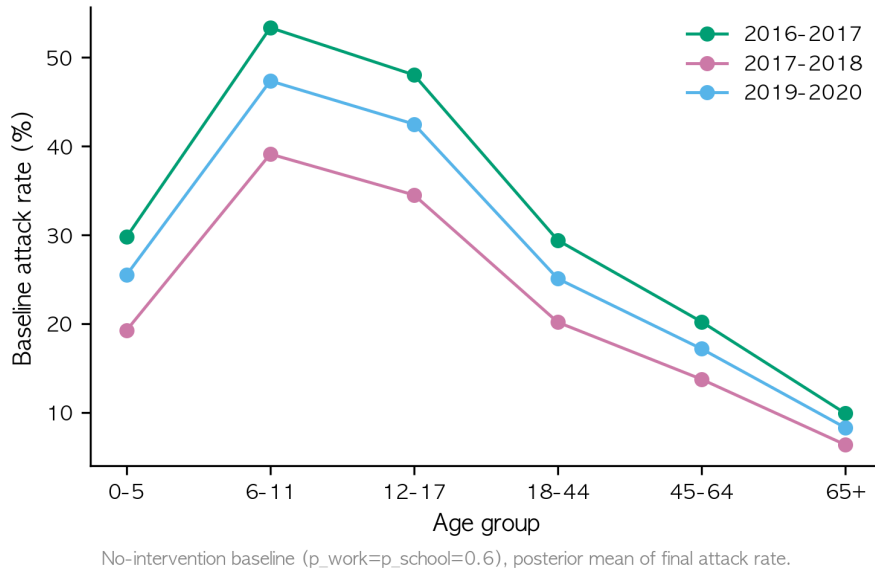
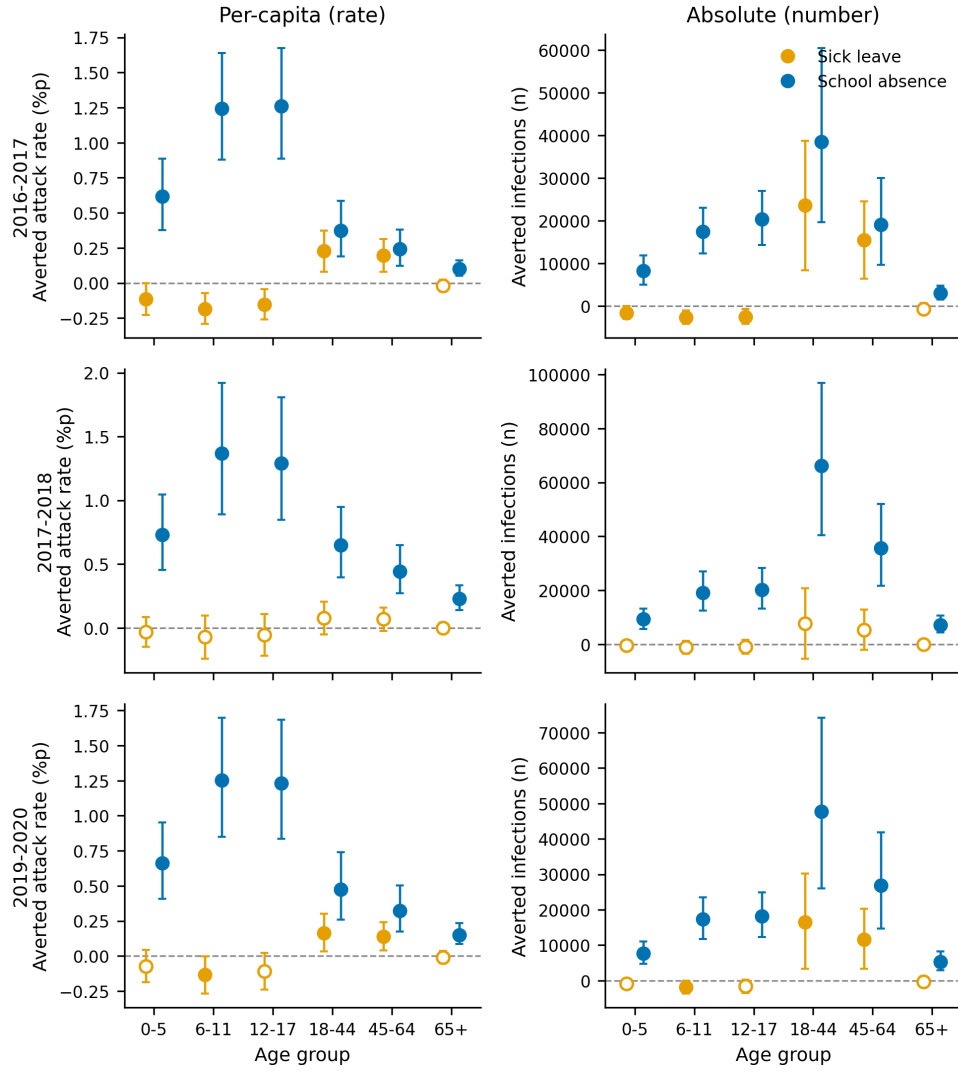


Figure S4: Baseline final attack rate by age band.

## S5 Age-specific policy effects across seasons

The main text shows the by-age effect in two metrics for the representative 2019–20 season. Figure S5 extends this to all three seasons, in per-capita (attack rate) and absolute (count) terms. The rate panels show that sick leave’s child-ward redistribution is clearest and most significant in the sharp 2016–17 season, weaker and non-significant in the mild 2017–18 season, and intermediate in 2019–20; the count panels show that the adult bands dominate the averted totals in every season.



Positive = averted. Filled =  $0 \notin \text{CI}$ , open =  $0 \in \text{CI}$ . Per-capita, sick leave lowers adults and raises children; in counts, adult bands dominate.

Figure S5: By-age averted effect of sick leave and school absence, all three seasons, in per-capita (left) and absolute (right) terms. Positive = averted; filled markers  $0 \notin \text{CI}$ , open markers  $0 \in \text{CI}$ .

## S6 Policy dose–response

Figure S6 shows total averted infection as a function of policy intensity for sick leave and school absence, at the identified crowding scale. School absence dominates across the full intensity range.

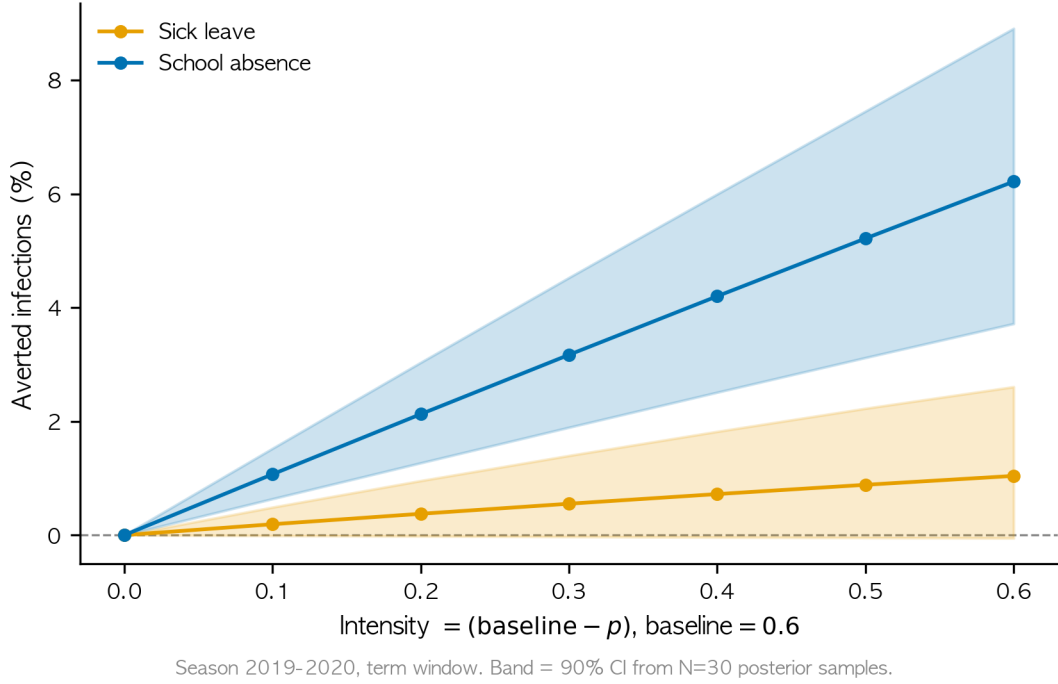


Figure S6: Total averted infection versus policy intensity for sick leave and school absence.

## S7 Additional sensitivity analyses

Figures S7–S9 extend the sensitivity battery summarized in the main text. Figure S7 shows how sick leave’s averted effect and the age redistribution vary with the crowding coefficient  $\kappa$ ; the net effect crosses zero near  $\kappa \approx 0.51$ , above the empirically identified range. Figure S8 shows that the child-ward winter break transfer does not reverse sign across the adult-immunity /  $R_0$  grid. Figure S9 decomposes the sick-leave effect by age across the school-term and winter-break windows, showing that the redistribution toward children intensifies during the winter break without reversing direction.

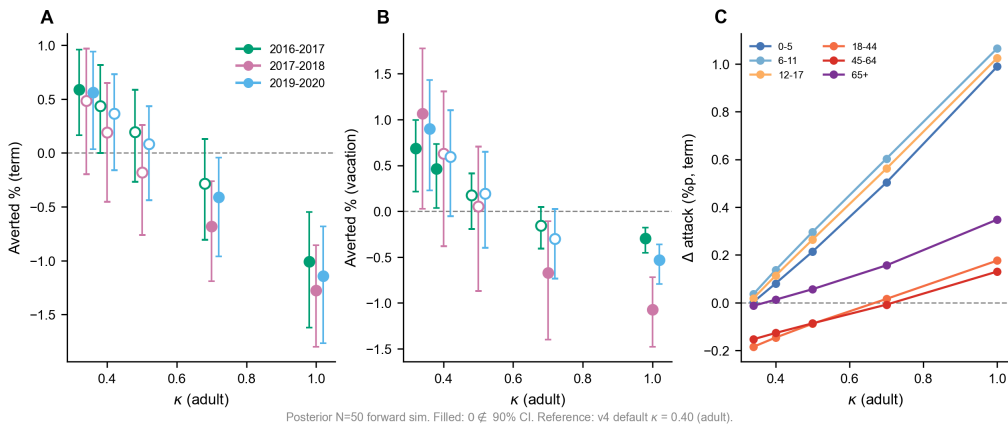


Figure S7: Sensitivity to the crowding coefficient  $\kappa$ : averted infection and age redistribution across the  $\kappa$  sweep, with credible intervals.

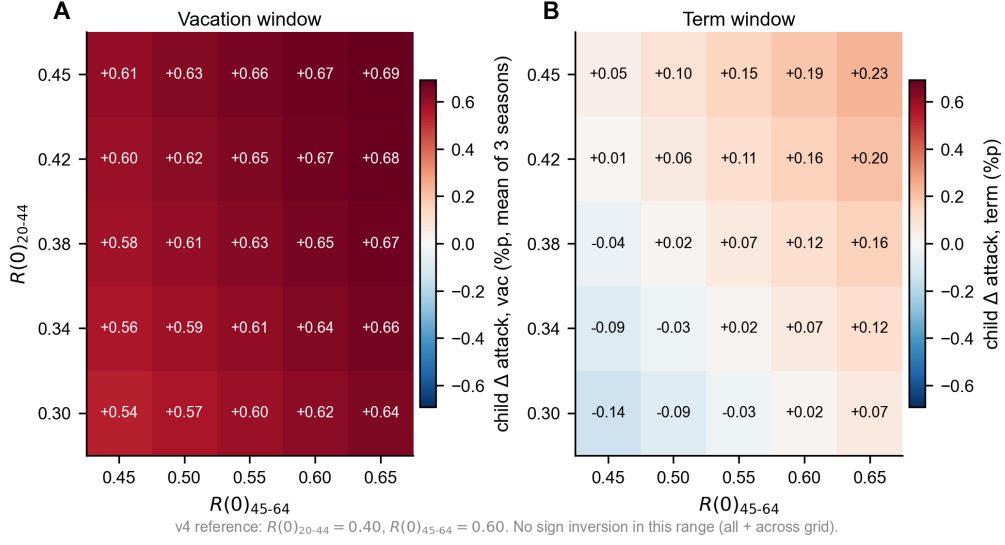


Figure S8: Sensitivity to adult initial immunity and  $R_0$ : sign of the child-ward winter break transfer across the parameter grid.

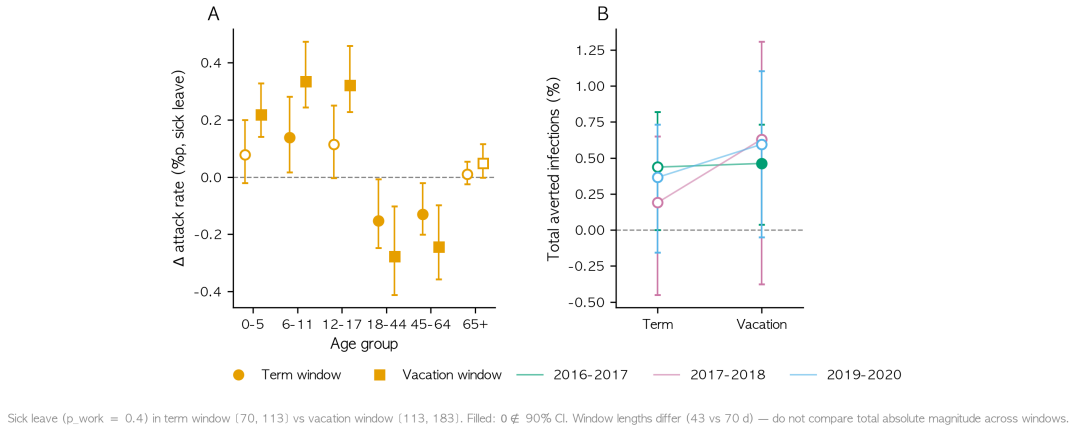


Figure S9: Sick-leave effect by age in the school-term versus winter-break windows (all age groups), with credible intervals.

## S8 Reproduction-number decomposition (all seasons)

This section complements the main-text reproduction-number decomposition (Section ??, shown for 2019–20) with the remaining two seasons and the underlying values. Because the channel allocation  $\pi$  is shared across seasons, the channel and age *shares* are essentially identical across seasons; only the absolute scale ( $R_0$ ) differs. Table S3 lists the decomposition, and Figures S10–S11 show  $R_e(t)$  for 2016–17 and 2017–18.

Table S3: Reproduction-number decomposition (posterior mean [90% CI]). Channel shares  $R_c/R_0$  with the amplification ratio relative to the transmissibility share  $\pi_c$ ; age shares  $R_b/R_0$ . Shares are shared across seasons ( $\pi$  common); the KDCA epidemic-advisory dates are season-specific.

| <i>Channel</i> ( $R_c/R_0$ , %)                 |                   | $R_c/\pi_c$     |
|-------------------------------------------------|-------------------|-----------------|
| home                                            | 44.6 [36.8, 51.9] | $1.0\times$     |
| work                                            | 3.8 [3.0, 4.6]    | $0.13\times$    |
| school                                          | 24.9 [21.2, 29.1] | $3.9\times$     |
| other                                           | 26.6 [21.7, 32.3] | $1.4\times$     |
| <i>Age</i> ( $R_b/R_0$ , %)                     |                   |                 |
| 0–5                                             |                   | 52              |
| 6–11                                            |                   | 18              |
| 12–17                                           |                   | 7               |
| 18–44                                           |                   | 17              |
| 45–64                                           |                   | 6               |
| 65+                                             |                   | 1               |
| <i>KDCA epidemic advisory (issued / lifted)</i> |                   |                 |
| 2016–17                                         |                   | 8 Dec / 2 Jun   |
| 2017–18                                         |                   | 1 Dec / 25 May  |
| 2019–20                                         |                   | 15 Nov / 27 Mar |

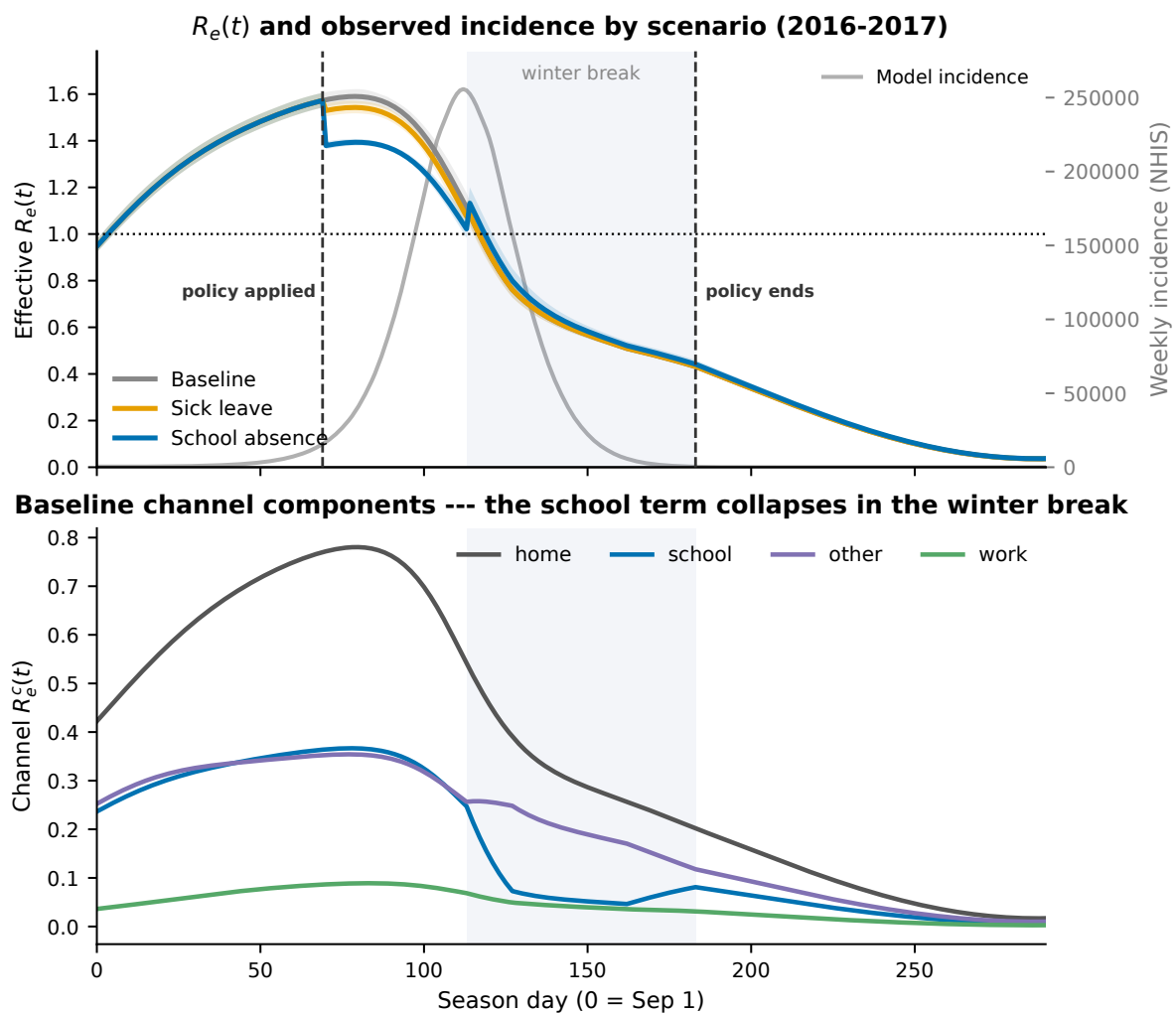


Figure S10: Effective reproduction number  $R_e(t)$  by scenario, 2016–17, with model incidence (right axis), the policy application window (dashed lines), and the baseline channel components of  $R_e(t)$ .

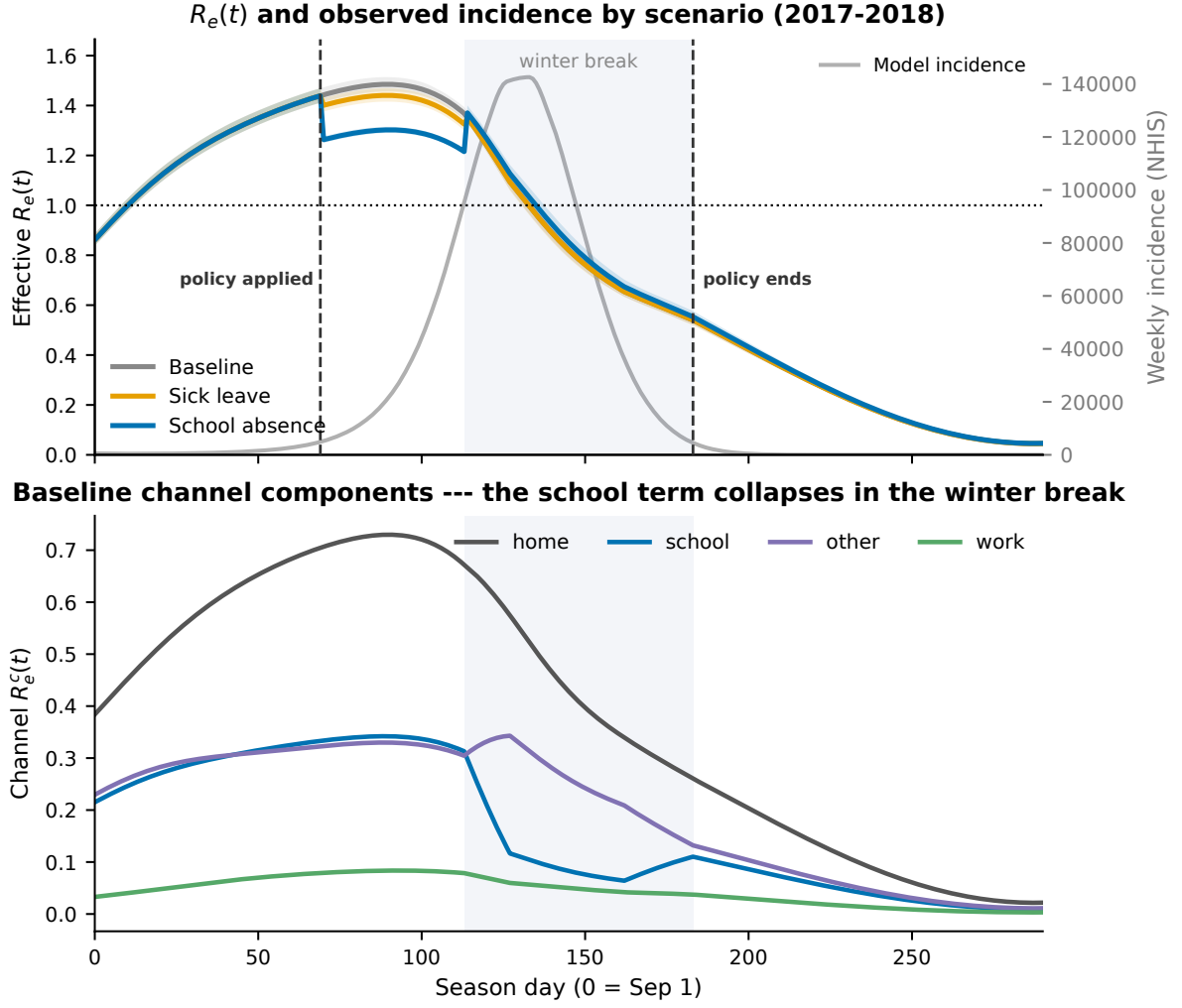


Figure S11: Effective reproduction number  $R_e(t)$  by scenario, 2017–18, with model incidence (right axis), the policy application window (dashed lines), and the baseline channel components of  $R_e(t)$ .

Figure S12 decomposes the policy effect by channel:  $\Delta R_e^c(t) = R_e^c(t)_{\text{policy}} - R_e^c(t)_{\text{baseline}}$  (2019–20). Each policy suppresses its own channel — school absence the school channel (by up to  $\approx 0.24$  during the school term) and sick leave the workplace channel (by up to  $\approx 0.06$ ) — with a small compensating rebound in the other channels (the redistribution). During the winter break the school channel has already collapsed at baseline, so school absence has little left to act on, consistent with the timing analysis in the main text.

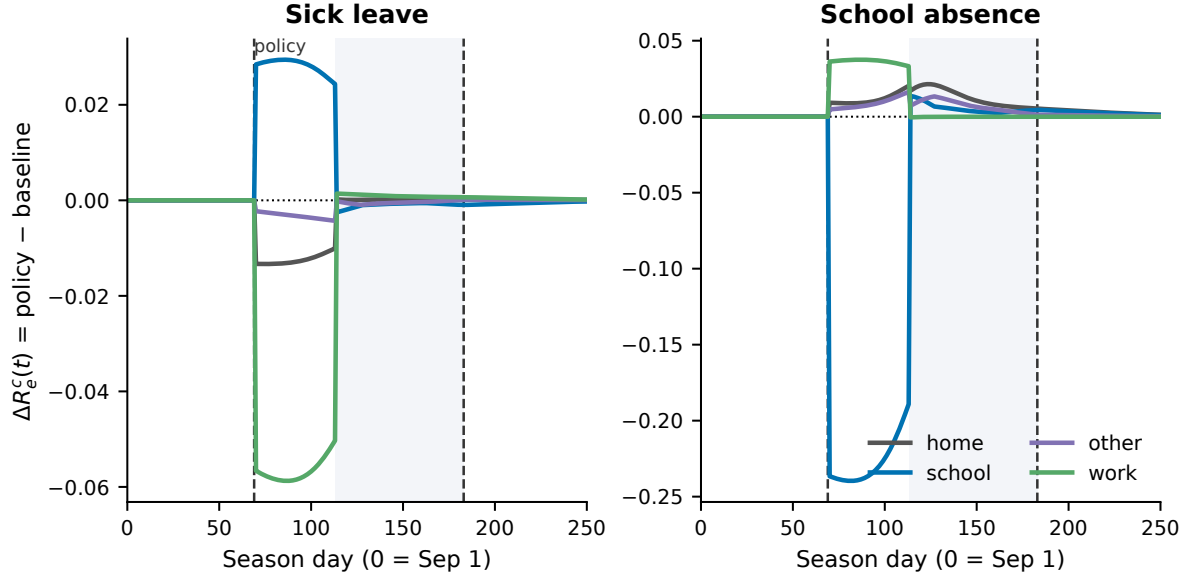


Figure S12: Policy-induced change in each channel's effective reproduction number,  $\Delta R_e^c(t) = \text{policy} - \text{baseline}$  (2019–20), for sick leave (left) and school absence (right). Dashed lines mark the policy application window, the shaded band the winter break. Note the different vertical scales.